

Algebra Practice: The Two Moves That Rearrange Any Equation

SYNOPSIS

Math is an integral part of doing physics at any level — we handle equations and algebraic expressions constantly. And you are already good at this, even if you don't know it. Imagine you're driving on a highway at 60 mph and your destination is 120 miles away. How long will it take? You immediately say **2 hours** — and to get that answer, you performed an algebraic manipulation in your head. In this note we slow that reflex down, name the two moves it uses — **cross over** (for multiplication and division) and **throw over** (for addition and subtraction) — and practice them until rearranging any physics equation is as automatic as the road-trip answer.

PERFORMANCE EXPECTATIONS After working through this note, you should be able to...

- 1 Distinguish terms from factors** in any equation: terms are separated by + or – signs; symbols multiplied or divided together form a single term made of factors.
- 2 Use the cross-over move:** when a factor crosses the equal sign, upstairs (numerator) becomes downstairs (denominator) and vice versa.
- 3 Use the throw-over move:** when a term is thrown over to the other side of the equal sign, its sign flips — positive becomes negative and vice versa.
- 4 Follow the two-step sequence** to solve for any symbol: first isolate the *term* that contains it (throw everything else over), then *bracket* the opposite side and divide out the unwanted *factor*.

- 5 **Recognize** the real physics equations on the Algebra Review Sheet — $v = f\lambda$, $f = 1/T$, $P = E/t$, $V = V_0 + 0.610t$, $I = I_g/4\pi R^2$ — as equations you will rearrange all semester. (You'll meet their *physics* — waves, sound, and light — as the course unfolds; here we only care about the algebra.)



equal intervals · constant speed



You already know this

You do algebraic manipulation in your head every day. You just haven't been calling it that.



The road trip

120 miles at 60 mph $\rightarrow$ 2 hours. You took distance and *divided by speed* — you rearranged $s = d/t$ into $t = d/s$ without writing anything down.



The headcount

A class of 25 has 10 males. Females? 15 — you did $25 - 10$. You rearranged $t = m + f$ into $f = t - m$ instantly.



Splitting the bill

A \$60 dinner among 4 friends is \$15 each. Total = (share) $\times$ (people), and you solved for the share: share = total/people. Cross-over, again.

Guiding questions

- I. What is the difference between a *term* and a *factor*?
- II. When a factor crosses the equal sign, what changes about its position?
- III. When a term is thrown over the equal sign, what changes about its sign?
- IV. In what order should you apply the two moves to isolate a chosen symbol — and what must you do *before* dividing out a factor?



Move 1 — Cross over (multiplication and division)

Remember that speed is the distance covered divided by the time taken:

$$\text{Speed (miles/hour)} = \frac{\text{Distance (miles)}}{\text{Time (hours)}}$$

To calculate the time taken on the road trip, you took the total distance of 120 miles and divided it by the speed, 60 miles per hour:

$$\text{Time (hours)} = \frac{120 \text{ (miles)}}{60 \text{ (miles/hour)}} = 2 \text{ hours}$$

Now use symbols — Speed = s , Distance = d , Time = t — and write both versions side by side:

$$s = \frac{d}{t} \quad \text{and} \quad t = \frac{d}{s}$$

Notice what changed from the first expression to the second: s in the numerator ("upstairs") became the denominator ("downstairs"), and t in the denominator ("downstairs") became the numerator ("upstairs") when s and t **crossed over** to the other side of the equal sign. To see it clearly, write every side as a fraction — any number divided by 1 is itself, so $s = s/1$:

$$\frac{s}{1} = \frac{d}{t} \quad \longrightarrow \quad \frac{t}{1} = \frac{d}{s}$$

That manipulation involved two simultaneous crossings. But it doesn't have to. Cross *only* t over and leave s where it is:

$$\frac{s}{1} = \frac{d}{t} \quad \longrightarrow \quad \frac{st}{1} = \frac{d}{1} \quad \longrightarrow \quad d = st$$

You just derived the distance covered when you travel at speed s for time t . Check it with the road trip: $(60 \text{ miles/hour}) \times (2 \text{ hours}) = 120 \text{ miles}$. ✓

This "trick" — **when you cross over the equal sign, you go from downstairs to upstairs and vice versa** — will *always* work, because it shortcuts the many steps of multiplying or dividing both sides that you're used to doing. Convince yourself once that the middle-school procedure gives the same answer; then don't bother with the middle-school procedure again.

Move 2 — Throw over (addition and subtraction)

There is a similar shortcut for addition and subtraction, and again you already do it in your head. A class of 25 has 10 males; you know instantly there are 15 females. What manipulation did you do? Let total = t , males = m , females = f , so $t = m + f$, that is, $25 = 10 + f$. To find f you did $25 - 10 = 15$. Algebraically: $f = t - m$.

The trick is this: **if you take a term and throw it over to the other side of the equal sign, its sign flips.** (A number without an explicit sign in front is positive.)

START

$$(+t) = (+m) + (+f)$$

THROW OVER

$$(+t) - m = (+f)$$

m crossed the equal sign; $+$ became $-$

Now let's do something weird. Throw t itself over to the right side:

START

$$(+t) = (+m) + (+f)$$

THROW OVER

$$0 = (+m) + f - t$$

you **must write a zero** when a side has no terms left — never leave it blank

CHECK

$$0 = 10 + 15 - 25 \checkmark$$

it makes total sense with the class numbers

The key message: **throw a term over, and positive becomes negative and vice versa.**

Putting both moves together

Now try an equation with both addition/subtraction *and* multiplication/division. This one comes straight from your Algebra Review Sheet — it gives the speed of sound V in air at Celsius temperature t (the physics comes later in the course; today it's just algebra):

$$V = V_0 + 0.610t$$

Let's solve for t (also said as "isolate t " or "make t the subject"). First, count the **terms** — pieces separated by $+$ or $-$ signs. The right side has two terms (V_0 and $0.610t$); the left side has one (V). Symbols that are multiplied or divided together are grouped as a *single* term: $0.610t$ is one term consisting of the **factors** 0.610 and t . Here is the sequence of operations you ought to follow:

START

$$V = V_0 + 0.610t$$

1·THROW OVER

$$-V_0 + V = +0.610t$$

isolate the **term** containing t : throw V_0 over and switch its sign

2·BRACKET

$$(-V_0 + V) = 0.610t$$

before you divide out, bracket together all terms on the opposite side

3·CROSS OVER

$$\frac{(-V_0 + V)}{0.610} = t$$

divide out the **factor** you don't want: 0.610 crosses over, upstairs $\rightarrow$ downstairs

TIDY

$$t = \frac{V - V_0}{0.610}$$

the same expression, written more neatly — this is Question 5 on the sheet

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Practice set — do these before class

Solve each equation for the requested symbol. Work them on paper first — the moves only become automatic through your hand, not your eyes. Bring your written work to class; we will go over these together and check the answers on the course page.

PROBLEM 1

Solve for a : $c = \frac{b}{a}$

PROBLEM 2

Solve for T : $f = \frac{1}{T}$

PROBLEM 3Solve for λ : $v = f\lambda$ **PROBLEM 4**Solve for E : $P = \frac{E}{t}$ **PROBLEM 5**Solve for y : $z = 3x + y$ **PROBLEM 6**Solve for x : $z = 3x + y$ **PROBLEM 7 • CHALLENGE**Solve for z : $\frac{c}{u} = \frac{z}{w}$ **PROBLEM 8 • CHALLENGE**Solve for R : $I = \frac{c}{R^2}$

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Summary: which move, when?

YOU SEE...	THE MOVE	WHAT HAPPENS
Symbols joined by $\times$ or $\div$ (factors in one term)	Cross over	Crossing the equal sign sends upstairs $\rightarrow$ downstairs and downstairs $\rightarrow$ upstairs. Bracket all terms on the opposite side first.
Pieces separated by $+$ or $-$ (terms)	Throw over	Throwing a term over the equal sign flips its sign: $+$ becomes $-$ and vice versa.
A side with no terms left	Write 0	You must write a zero — an empty side is never left blank.

Looking ahead: after we review these in class, you'll take the **Algebra Review Quiz** on Brightspace. Its questions mirror the practice set above — and the **Algebra Review Sheet** handout has the ten real physics equations the quiz draws from: wave speed, period-frequency, the Doppler pair, the speed of sound, power, and the inverse-square law. Don't worry about what the symbols mean yet — we'll meet the physics as the semester unfolds. Everything on the sheet uses nothing beyond the two moves in this

note, except one square root at the very end of Question 9. Ask for help with any you don't remember how to do.

Three rules to remember



$$s = d/t \Leftrightarrow st = d$$

Cross over. A factor crossing the equal sign goes from downstairs to upstairs and vice versa.



$$t = m + f \Leftrightarrow t - m = f$$

Throw over. A term thrown across the equal sign flips its sign.



$$(V - V_0) = 0.610t$$

Bracket first. Before dividing out a factor, bracket together all terms on the opposite side.

EXIT TICKET · CHECK YOURSELF BEFORE CLASS

Q1. Solve $V = V_0 + 0.610t$ for V_0 .

☐ $V_0 = V - 0.610t$

☐ $V_0 = V + 0.610t$

☐ $V_0 = 0.610t - V$

☐ $V_0 = V / (0.610t)$

Q2. Solve $v = f\lambda$ for f .

☐ $f = v\lambda$

☐ $f = \lambda/v$

☐ $f = v/\lambda$

☐ $f = v - \lambda$

Q3. Using your result $f = v/\lambda$ with $v = 340$ m/s and $\lambda = 2.0$ m, calculate f .

Hz

Q4. You throw the term V_0 from the right side of $V = V_0 + 0.610t$ over to the left side.

What happens to it?

☐ It goes downstairs (becomes a denominator).

☐ Its sign flips: it arrives as $-V_0$.

☐ Nothing changes; terms move freely.

☐ It disappears, leaving a zero in its place.